

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Economics

Semester: Spring

Year : 2024
Full Marks:100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Write the key principles of economy. 5
b) Explain the Manufacturing cost and non-manufacturing cost. 5
c) Describe the ecological foot print. 5
2. a) A man aged 30 years now had borrowed Rs. 5,00,000 from a bank for his further studies at the age of 20 years. The bank was charged interest at 12% per year compounded quarterly. He wishes to pay that loan from last 10 semi annual way with equal installment basis and now he has just cleared the loan. What amount did he pay in each installment? 7
b) Star construction company planning to purchase from different two companies A & B. Because it was based on estimation using co-Terminated 5 years and 8 years respectively. 8

	Investment Rs.	Revenue Rs.	Expenses Rs.	Salvage value Rs.
Plain A	350000000	13000000	1500000	3500000
Plain B	500000000	17500000	2500000	5000000

3. a) Find both types of B/C ratios using PW formulation for a project having first investment cost of Rs. 1,50,000; project life 10 years; salvage value Rs. 30,000; annual benefit Rs. 65,000; annual O & M cost Rs. 20,000 & MARR=12% per year. 7
b) Perform sensitivity analysis of the following project over the range of $\pm 30\%$ in i) Initial investment ii) Annual revenue iii) Useful life. 8

Initial investment = 14,500

Annual revenue = 4,000

Salvage value = 1,000

Useful life = 6 years

MARR = 20%

4. a) Compute discounting payback period and B/C ratio from the following data. 7
- Initial investment Rs. 4, 00,000
 Annual revenue Rs. 1,50,000
 Annual cost Rs. 30,000
 Salvage value Rs. 50,000
 Useful life year 8
 MARR 12%
- b) Select the best project by using IRR method when MARR is 8%. 8
- | Items | Project A | Project B |
|--------------------|-----------|-----------|
| Initial investment | 3,00,000 | 5,00,000 |
| Annual revenue | 90,000 | 1,75,000 |
| Life Year | 6 | 6 |
| Salvage value | 10,000 | 1,00,000 |
5. a) After passing civil Engineering course suppose you are going to start housing construction work, If you started it, describe the project funding mechanism, How many funding mechanism are there please describe. 8
- b) What are financial ratios and why are they important in financial analysis and decision making? How do financial ratios provide insight into a company's performance and financial health? 7
6. a) A construction equipment has initial cost & annual saving per year are of Rs 40000 & Rs 20000 respectively with annual operation & maintenance cost of 7000. It will depreciate by MACRS method & will have no salvage value. The useful life of equipment is 5 years. Estimate before & after tax cash flow. The company pays income tax @40%. Evaluate after tax by FW method. 8
- b) Describe the Taxation law and what do you understand about corporate income Tax? 7
7. Write short notes on: (**Any two**) 2×5
- Balance sheet
 - Nominal rate of interest
 - Demand and supply